

AI-Powered C Programming Tutor - Setup Guide

Welcome to the **AI-Powered C Programming Tutor**, an intelligent educational platform designed to teach C programming using advanced AI technologies.

This system leverages **RAG (Retrieval-Augmented Generation)**, a **Vector Database (ChromaDB)**, and a **Knowledge Graph (Neo4j)** to provide a comprehensive learning experience. It features distinct interfaces for **Students** (Active Learning) and **Teachers** (Class Management).

Prerequisites

Before running the system, ensure you have the following installed on your machine:

- **Python 3.10** or higher
- [Neo4j Desktop](#) (Required for the Knowledge Graph)
- [Google Gemini API Key](#)

Installation & Setup

Step 1: Environment Setup

1. **Clone/Unzip** the project to your local machine.
2. Open a **Terminal** inside the project folder.
3. **Create a Virtual Environment** (Recommended):

Windows:

```
python -m venv venv  
venv\Scripts\activate
```

Mac/Linux:

```
python -m venv venv  
source venv/bin/activate
```

4. **Install Dependencies:**

First, ensure you have a requirements.txt file. If not, create one with the necessary packages or run the command directly:

```
pip install fastapi uvicorn google-generativeai langchain-text-splitters chromadb neo4j  
python-dotenv python-multipart
```

Step 2: Database Configuration

1. Setup Neo4j (Knowledge Graph)

1. Open **Neo4j Desktop**.
2. Click **New Project** -> **Add Local DBMS**.
3. Set the password to: password (or update your .env file later).
4. **Start** the Database.
5. (Optional) Install the **APOC plugin** if you plan to use advanced graph features.

2. Configure Environment Variables

Create a file named .env in the root directory of the project and paste the following configuration:

```
# API Keys
GOOGLE_API_KEY=your_actual_google_api_key_here
```

```
# Neo4j Settings
NEO4J_URI=bolt://localhost:7687
NEO4J_USER=neo4j
NEO4J_PASSWORD=password
```

```
# RAG Settings
# STEM_CHUNK_SIZE=1200
```

Step 3: Data Ingestion (Building the Brain)

You must populate the databases before the bot can answer questions.

1. **Run Automatic Ingestion** (Reads .c and .md files):
python scripts/ingest_data.py

Success Message: "Processed X total chunks."
2. **Run Manual Graph Ingestion** (Builds logic connections):
python scripts/ingest_manual_graph.py

Success Message: "Manual Graph Ingestion Complete."

Step 4: Launching the System

1. **Start the Backend Server:**
uvicorn app.main:app --reload

Verify output: Uvicorn running on http://127.0.0.1:8000
2. **Open the Frontend:**
 - Navigate to your project folder in your file explorer.

- Double-click index.html to open it in your web browser.

User Guide & Test Scenarios

The system uses a login based on database/users.csv.

Student Mode

Login Credentials:

- **Username:** student1 (or student_hacker)
- **Password:** pass123

Features to Test:

- **Concept Q&A:** Ask "What is an array?" Verify you see a text explanation **AND** a Flowchart/Diagram.
- **Active Coding:** Ask "How do I declare a variable?" Wait for the buttons, then click "💪 Challenge". Paste code (e.g., int x = 10;) and verify the bot gives feedback.
- **Dashboard:** Click " Dashboard" in the header to view learning stats.

Teacher Mode

Login Credentials:

- **Username:** teacher1
- **Password:** admin123

Features to Test:

- **Curriculum Control:**
 1. Toggle "**Arrays**" to **OFF** on the Dashboard.
 2. Go to Chat (as Student) and ask about Arrays. The bot should refuse to answer.
- **Analytics:** Check "Class Analytics" charts to see student activity.
- **Resource Upload:** Use the Upload section to add a new .md file. The system will auto-update.
- **Exam Access:** Go to Chat (as Teacher) and ask "What is the solution to Question 1?". The bot will reveal the answer key (requires exam_solutions.md).

Troubleshooting

Issue	Solution
"Connection Error" in Chat	1. Ensure backend is running (uvicorn). 2. Check GOOGLE_API_KEY in .env.
"Database Empty" Error	1. Run python scripts/debug_retrieval.py to

	check status. 2. Re-run ingestion scripts in Step 3.
Neo4j Errors	1. Ensure Neo4j database is Started (Green). 2. Verify password matches .env.